

MD MAZHARUL ISLAM EMON

Full-Stack AI Engineer

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SUMMARY

Full-Stack AI Engineer with 3+ years in software engineering — building scalable backends, cross-platform mobile apps, and AI-powered SaaS platforms. Ships production systems with Django, Flutter, and Next.js, and delivers faster using AI-native tooling like Claude Code and Cursor. Proven track record in multi-tenant architecture, real-time speech evaluation pipelines, and end-to-end product delivery across 93+ production releases. Strong background in competitive programming with 2,000+ problems solved and ICPC Asia Dhaka Regional participation.

TECHNICAL SKILLS

Languages & Frameworks: Python · Django · Django REST Framework · FastAPI · Dart · Flutter · TypeScript · Next.js · React
Databases & Messaging: PostgreSQL · Redis · Celery · Celery Beat
AI & NLP: Speech Recognition · NLP Processing · LLM Integration · PyTorch · WhisperX · OpenAI API
Infrastructure & DevOps: Docker · Nginx · GCP · DigitalOcean · CI/CD · Prometheus · Grafana · Firebase · FCM
AI-Assisted Development: Claude Code · Cursor · Agentic Workflows · AI Pair Programming

PROFESSIONAL EXPERIENCE

LII Lab Sylhet, Bangladesh
Senior Software Engineer 01/2025 – Present

Led backend engineering for OneIELTS and OnePTE — AI-powered English test preparation platforms.

- Architected **multi-tenant SaaS backend** with django-multitenant — full data isolation, custom domain routing, per-tenant scoring config and feature flags.
- Designed **QTI 3.0-compliant exam engine** supporting 100+ question types across IELTS Academic and General, with XML parsing, interaction routing, and strategy-pattern band-score normalization.
- Built **real-time speech evaluation pipeline** integrating Speech Recognition, Pronunciation Assessment, and NLP Processing — end-to-end scoring in under 15 seconds.
- Engineered **multi-gateway payment infrastructure** with 5 providers (Stripe, Razorpay, SSLCommerz, Google Play, Apple), subscription lifecycle management, and webhook idempotency.
- Built **Studio API layer** for moderator platform — content authoring, expert evaluation queue, exam rejudge pipelines, and KPI analytics for subscriptions and engagement.
- Set up **production observability stack** with Prometheus and Grafana — health monitoring, per-API latency instrumentation, and alerting.
- Mentored junior engineers through code reviews, PR feedback, and architectural walkthroughs.

Stack: Django · DRF · Python · PostgreSQL · Redis · Celery · FastAPI · Docker · GCP · Prometheus · Grafana · Next.js · TypeScript

LII Lab Sylhet, Bangladesh
Software Engineer 11/2022 – 12/2024

Full-stack ownership of OnePTE — designed and shipped the Flutter app from day one and built the Django AI scoring backend.

- Designed and built **Flutter cross-platform app** from scratch — all four PTE modules (Speaking, Writing, Reading, Listening) with 20+ task types, audio recording, and timer management.
- Built task-group-based **mock test engine** with modular exam templates, configurable time allocations, automated question progression, and multi-dimensional score breakdowns.
- Engineered **AI scoring backend** for spoken and written PTE tasks using Speech Recognition, Pronunciation Assessment, and NLP Processing for multi-trait evaluation.
- Integrated **subscription billing across 4 platforms** (Stripe, SSLCommerz, Google Play, Apple App Store) with webhook handling, transaction deduplication, and in-app purchase verification.
- Built **Django admin and private API** for content moderation, question bank management, subscription analytics with regional reporting, and user acquisition dashboards.
- Led **93+ production releases** across Android, iOS, and Web — managed release pipelines, Firebase config, and CI/CD across all channels.

Stack: Flutter · Dart · Riverpod · Django · DRF · Python · PostgreSQL · Redis · Celery · Firebase · Next.js · TypeScript

PROJECTS

Patty Bros — Restaurant Website

2025

- Built **responsive marketing and booking site** for a South East London restaurant — animated pages with Framer Motion and Zod-validated booking forms.
- Integrated **Google Sheets-driven live menu** with ISR revalidation, and a serverless booking backend via Google Apps Script with real-time Telegram notifications.

Stack: Next.js · TypeScript · Tailwind CSS · Framer Motion · Google Sheets API · Google Apps Script · Telegram Bot API

Projecto — University Course & Proposal Management

10/2022 – 12/2022

- Built **multi-role Django web app** (student, supervisor, admin) for university project proposal management with AJAX-driven interactions and automated email notifications.
- Implemented **PDF and CSV export** for project reports and integrated background job service for scheduled operations.

Stack: Django · Python · PostgreSQL · AJAX · jQuery · Bootstrap

COMPETITIVE PROGRAMMING & ACHIEVEMENTS

Solved **2,000+ problems** across online judges; participated in **300+ contests**.

Champion, LU CSE Carnival National Hackathon 2023 — Team: LU Ovream

Runner-up (Bangladesh), IEEEExtreme 16.0 (2022) — Global Rank 149, Team: LazySquad

ICPC: 87th / 1,700+ teams, Preliminary 2021; 51st, Dhaka Regional 2020

Champion, LU TechStorm 4 Programming Contest 2021

Codeforces (Max: **1603**) · CodeChef (Max: **1965**) · LightOJ

EDUCATION

Leading University

Sylhet, Bangladesh

B.Sc. in Computer Science and Engineering — GPA: 3.6 / 4.0

09/2018 – 12/2022

VOLUNTEERING

IEEE Computer Society — LU Student Branch Chapter

Chair

04/2022 – 05/2023

- Established the IEEE CS Student Branch Chapter at LU — organized seminars, webinars, and technical workshops with industry speakers.
- Grew chapter **membership by 20%+** through outreach programs; delivered a multi-session **Flutter bootcamp**.
- Served as problem setter and judge for inter-university programming contests.

Leading University Computer Club

ACM Coordinator

01/2022 – 12/2023

- Conducted peer-to-peer DSA sessions and mentored juniors in competitive programming strategies across Codeforces, CodeChef, and LightOJ.
- Organized regular contest practice sessions to prepare members for ICPC and national competitions.